ECON6067 Empirical Report

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1. Research Question and Motivation

Momentum investing is a widely discussed investment strategy in financial markets, including China's A-share market. It is based on the idea that stocks with strong past performance tend to continue performing well in the near future, while past losers tend to underperform. As an individual investor, I have frequently encountered momentum-based trading strategies in market discussions and investment analysis. This motivated me to develop a better understanding of whether momentum investing is supported by empirical evidence rather than relying solely on market experience.

Jegadeesh and Titman (1993), in their paper *Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency*, examine whether buying past winners and selling past losers can generate statistically significant positive returns. Their findings suggest that momentum strategies based on past 3- to 12-month returns can generate positive abnormal returns, challenging the traditional view of market efficiency.

The main research question of this project is: **Do the momentum strategies documented by Jegadeesh and Titman (1993) continue to generate positive and statistically significant returns in the modern US stock market?**

To address this question, this project replicates Tables I–III of the original paper using more recent US stock market data. It examines the profitability of different momentum strategies and investigates whether their returns can be explained by market risk or firm size. An independent extension further compares the replicated momentum strategy with the Fama–French Momentum Factor.

Although this study focuses on the US equity market rather than China's A-share market, it provides an empirical foundation for understanding momentum investing and its potential limitations in practical investment decisions.

2. Data Sources, Sample Construction, and Variable Definitions

2.1 Data Sources

This study uses stock return data from the Center for Research in Security Prices (CRSP), provided through the course datasets. Unlike the original study, which examines the period from 1965 to 1989, this project conducts a modern-sample replication using data covering January 2000 to December 2025.

Both monthly and daily stock data are used. Monthly data provide individual stock returns and market capitalization, while daily data are used for the one-week gap strategies and beta estimation.

The CRSP value-weighted market index is used to measure market returns, and the one-month risk-free rate is obtained from the Fama–French factor dataset.

For the independent extension, the monthly Momentum Factor (Mom) is obtained from the Kenneth French Data Library.

2.2 Sample Construction

Following the original paper, the sample is restricted to common stocks listed on the New York Stock Exchange (NYSE) and the American Stock Exchange (AMEX, now NYSE American).

The original monthly dataset contains 2,423,212 observations covering 25,301 distinct securities. Before constructing the portfolios, duplicate security-month records are resolved using historical security information and effective dates. The sample is then restricted to eligible US common stocks using CRSP exchange and security classification codes.

Specifically, stocks are retained if they satisfy the following criteria: (1) *PrimaryExch* is *N* or *A*, representing NYSE or NYSE American. (2) *SecurityType* is *EQTY* and *SecuritySubType* is *COM*.(3) *ShareType* is *NS* and *USIncFlg* is *Y*.(4) *IssuerType* is *ACOR* or *CORP*.

After resolving duplicates and applying these selection criteria, the prepared sample contains 520,438 stock-month observations representing 4,669 securities.

Missing returns are not automatically replaced with zero, and extreme returns are not removed solely because of their magnitude. Instead, observations are assessed according to their availability and suitability for portfolio construction. Valid delisting returns are retained where available to reduce survivorship bias.

The initial data cover January 2000 to December 2025. To ensure sufficient historical observations for portfolio formation and overlapping holding periods, the main Table I analysis uses a common evaluation period from January 2002 to December 2025, comprising 288 months.

2.3 Variable Definitions

The main variables used in the empirical analysis are defined as follows.

Table 2.3 Main Variable Definition

Variable	Definition
Stock Return ($R_{i,t}$)	Monthly total return of stock (i), including distributions where available.
Formation Return	Cumulative return over the preceding (J) months, used to rank stocks.
Holding Period (K)	Number of months for which a portfolio is held after formation.
Winner Portfolio (P10)	Equally weighted portfolio of stocks in the highest past-return decile.
Loser Portfolio (P1)	Equally weighted portfolio of stocks in the lowest past-return decile.

Momentum Return (WML)	Difference between winner and loser portfolio returns, $(R_{P10,t} - R_{P1,t})$.
Market Capitalization	Market value of a company's outstanding shares, used for size-related analysis.
Market Beta (β)	Sensitivity of stock or portfolio returns to market returns.
Risk-Free Rate (R_f)	Monthly one-month Treasury bill rate used in risk-adjusted analysis.
CAPM Alpha (α)	Abnormal return unexplained by exposure to the market risk factor.

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The main momentum strategy is constructed by ranking stocks according to their cumulative past returns:

$$R_{i,t}^{(J)} = \prod_{s=1}^{J} (1 + R_{i,t-s}) - 1$$

The momentum portfolio return is then calculated as:

$$R_{WML,t} = R_{P10,t} - R_{P1,t}$$

These variables form the basis of the momentum portfolio construction and subsequent risk-adjusted analysis.

3. Empirical Design

This study follows the portfolio-sorting methodology developed by Jegadeesh and Titman (1993) to examine the profitability of momentum strategies in the modern US stock market. The empirical analysis consists of three parts corresponding to Tables I–III of the original paper.

3.1 Table I: Momentum Strategy Returns

Stocks are ranked according to their cumulative returns over the previous J months, where J = 3, 6, 9, or 12. Based on these rankings, stocks are divided into ten equally weighted portfolios, with P1 representing past losers and P10 representing past winners.

A zero-cost momentum strategy is constructed by buying P10 and selling P1. Each portfolio is held for K = 3, 6, 9, or 12 months, with overlapping portfolios formed monthly. Two specifications are examined: Panel A forms portfolios without a gap, while Panel B introduces a one-week gap between the formation and holding periods to reduce potential short-term price effects.

The average monthly returns and t-statistics are calculated for 32 strategies. All strategies are evaluated over the common period from January 2002 to December 2025.

3.2 Table II: Market Risk and Firm Size

Following the original paper, the analysis focuses on the six-month formation and six-month holding strategy (J=6, K=6), without skipping a week.

Post-ranking market betas are estimated using regressions of portfolio returns on CRSP value-weighted market returns. Average market capitalization is also calculated for each momentum decile. These measures help determine whether momentum profitability can be explained by differences in systematic risk or firm size.

3.3 Table III: Size- and Beta-Based Momentum Strategies

To further investigate the sources of momentum profits, stocks are divided into three groups according to market capitalization and, separately, three groups according to their ex-ante market betas.

Within each subgroup, the 6/6 momentum strategy is implemented again. Ex-ante betas are estimated using the Scholes–Williams method based on daily stock returns from the preceding calendar year.

Panel A reports average monthly portfolio returns, while Panel B estimates risk-adjusted returns using the CAPM regression:

$$R_{p,t} - R_{f,t} = \alpha_p + \beta_p(R_{m,t} - R_{f,t}) + \varepsilon_{p,t}$$

where α_p represents the abnormal return after controlling for market risk. The analysis also includes joint tests of portfolio returns and alphas.

4. Core Replication Results and Comparison with the Original Paper

4.1 Table I: Momentum Strategy Returns

The replication results show that all 32 winner-minus-loser strategies generate positive average monthly returns over the modern sample period. This is directionally consistent with Jegadeesh and Titman (1993), who documented positive momentum profits in the original 1965–1989 sample.

However, the magnitude and statistical significance of momentum returns are generally weaker in the modern sample.For the benchmark 6/6 strategy without a one-week gap, the replicated winner-minus-loser portfolio earns an average monthly return of 0.659%, compared with 0.95% in the original paper. Its t-statistic is 1.51, compared with 3.07 in the original study.

When a one-week gap is introduced, the replicated 6/6 return increases to 0.726% per month, compared with the original return of 1.10%. Although the one-week gap slightly improves

performance, the replicated return remains statistically insignificant at the conventional two-sided 5% level.

Among all strategies, the highest modern average monthly WML return is 0.879%, achieved by the 9/3 strategy with a one-week gap. Nevertheless, none of the 32 strategies reaches the conventional two-sided 5% significance threshold.

Overall, the results suggest that momentum-related return continuation remains observable in the modern sample, but the evidence for statistically significant raw momentum profits is weaker than in the original study.

Table 4.1 Momentum Strategy Returns: Replication vs. Original Study

Panel	J/K	Modern WML (%)	Modern t	Original WML (%)	Original t
A	3/3	0.336	0.93	0.320	1.10
A	6/3	0.627	1.38	0.840	2.44
A	6/6	0.659	1.51	0.950	3.07
A	6/12	0.416	1.16	0.860	3.36
A	9/3	0.818	1.58	1.090	3.03
A	12/12	0.148	0.35	0.680	2.25
B	3/3	0.491	1.38	0.730	2.61
B	6/3	0.740	1.64	1.140	3.37
B	6/6	0.726	1.68	1.100	3.61
B	6/12	0.440	1.24	0.900	3.54
B	9/3	0.879	1.71	1.350	3.85
B	12/12	0.163	0.38	0.690	2.31

WML mean monthly return in percent; t-statistics in separate columns. Panel A: no skip; Panel B: one-week skip. N = 288 months (modern sample).

4.2 Table II: Betas and Market Capitalization

Table II examines whether the returns of momentum portfolios are related to systematic market risk and firm size.

The replicated results show that the loser portfolio (P1) has a post-ranking market beta of 1.889, while the winner portfolio (P10) has a beta of 1.255. The resulting winner-minus-loser beta is therefore −0.634, compared with −0.08 in the original paper.

This indicates that past losers exhibit greater market sensitivity than past winners in the modern sample. Therefore, the positive average return of the momentum strategy cannot be explained simply by greater positive exposure to market risk.

In terms of firm size, the average market capitalization of P1 is approximately $1.80 billion, compared with $5.64 billion for P10. This suggests that past losers tend to be smaller firms than past winners, consistent with the general pattern documented in the original study.

However, the capitalization figures are not directly comparable across the two sample periods because of differences in nominal prices, inflation, and the composition of listed firms.

Table 4.2 Post-ranking Market Beta and Firm Size by Momentum Decile

Decile	Modern β	Original β	Modern Cap ($m)	Original Cap ($m)
P1	1.889	1.36	1,799	208
P2	1.483	1.19	5,438	480
P3	1.311	1.14	9,265	545
P4	1.194	1.11	11,261	619
P5	1.104	1.09	12,632	693
P6	1.044	1.08	13,652	703
P7	1.010	1.09	14,182	738
P8	1.046	1.12	13,323	759
P9	1.088	1.17	11,317	680
P10	1.255	1.28	5,642	495
P10-P1	-0.634	-0.08		

J=6, K=6, no skip. Market capitalization in USD millions, nominal amounts in different time periods; values are not inflation-adjusted.

4.3 Table III: Size- and Beta-Based Momentum Strategies

Table III examines whether momentum profits remain after stocks are grouped according to size and market beta.

For the size-based portfolios, the replicated average monthly WML returns are 0.957%, 0.583%, and 0.316% for small, medium, and large firms, respectively. This suggests that momentum profitability is stronger among smaller firms and weaker among large firms.

For the beta-based portfolios, the average monthly WML returns are 0.380%, 0.415%, and 1.223% for low-, medium-, and high-beta stocks, respectively. The high-beta subgroup generates the strongest return, with a t-statistic of 2.07.

After controlling for market risk using the CAPM, the full-sample winner-minus-loser portfolio produces a monthly alpha of 1.118%, with a t-statistic of 2.72. This is statistically significant at the conventional 5% level, despite the weaker significance of the unadjusted return.

The negative market beta of the momentum strategy helps explain why its CAPM-adjusted alpha is higher than its raw average return.

The risk-adjusted results also show significant positive alphas for the small- and medium-size groups and for the medium- and high-beta groups at the conventional 5% level.

Overall, Table III provides additional evidence that momentum performance cannot be explained solely by exposure to market risk, although its strength varies across different stock characteristics.

Table 4.3 Momentum Returns and CAPM Alphas Across Size and Beta Groups

Group	Raw (%)	Raw t	Orig. Raw (%)	Alpha (%)	Alpha t	Orig. Alpha (%)
All	0.659	1.51	0.950	1.118	2.72	1.000
S1	0.957	1.67	0.990	1.501	2.75	1.060
S2	0.583	1.39	1.260	0.972	2.43	1.290
S3	0.316	0.98	0.750	0.592	1.89	0.760
B1	0.380	1.05	0.620	0.664	1.89	0.630
B2	0.415	1.15	0.790	0.750	2.17	0.830
B3	1.223	2.07	1.080	1.767	3.13	1.110

Percent per month. Panel A reports raw WML returns; Panel B reports CAPM WML alphas. S1–S3: small/medium/large; B1–B3: low/medium/high ex-ante beta.

5. Independent Extension: Comparing JT Momentum with the Fama–French Momentum Factor

As an independent extension, this study compares the replicated Jegadeesh–Titman 6/6 momentum strategy with the Momentum Factor (Mom) from the Kenneth French Data Library.

Although both strategies are based on past stock returns, their construction methods differ. The replicated JT strategy uses equally weighted winner and loser decile portfolios, whereas the French Mom factor incorporates size-based sorting and value-weighted portfolios.

The comparison uses 288 monthly observations from January 2002 to December 2025.The replicated JT strategy generates an average monthly return of 0.659%, compared with 0.134% for the French Mom factor. Their monthly standard deviations are 7.425% and 4.483%, respectively. This indicates that JT momentum generates a higher average return but also experiences greater volatility.

The Pearson correlation coefficient between the two strategies is 0.821, suggesting a strong positive relationship.To further investigate their relationship, the following time-series regression is estimated:

$$R_{JT,t} = \alpha + \beta MOM_t + \varepsilon_t$$

The estimated beta is 1.360, indicating that the replicated JT strategy has relatively strong exposure to the French momentum factor. The regression R-squared is 0.675, meaning that approximately 67.5% of the variation in JT monthly returns is explained by the linear relationship with Mom.

The estimated monthly alpha is 0.476%, with a t-statistic of 1.90 and a p-value of 0.058. This alpha is not significant at the 5% level, although it is significant at the 10% level.

These results suggest that the two strategies capture closely related momentum patterns, while differences in sorting periods, weighting methods, and stock universes may account for some of their performance differences.

Table 5.1 Independent Extension: JT WML vs. French Momentum Factor

Statistic	JT WML	French Mom
Mean monthly return (%)	0.659	0.134
Monthly volatility (%)	7.425	4.483
Mean t-statistic	1.506	0.509
Positive months (%)	66.667	56.944
Minimum monthly return (%)	-49.527	-34.360
Maximum monthly return (%)	19.755	12.730

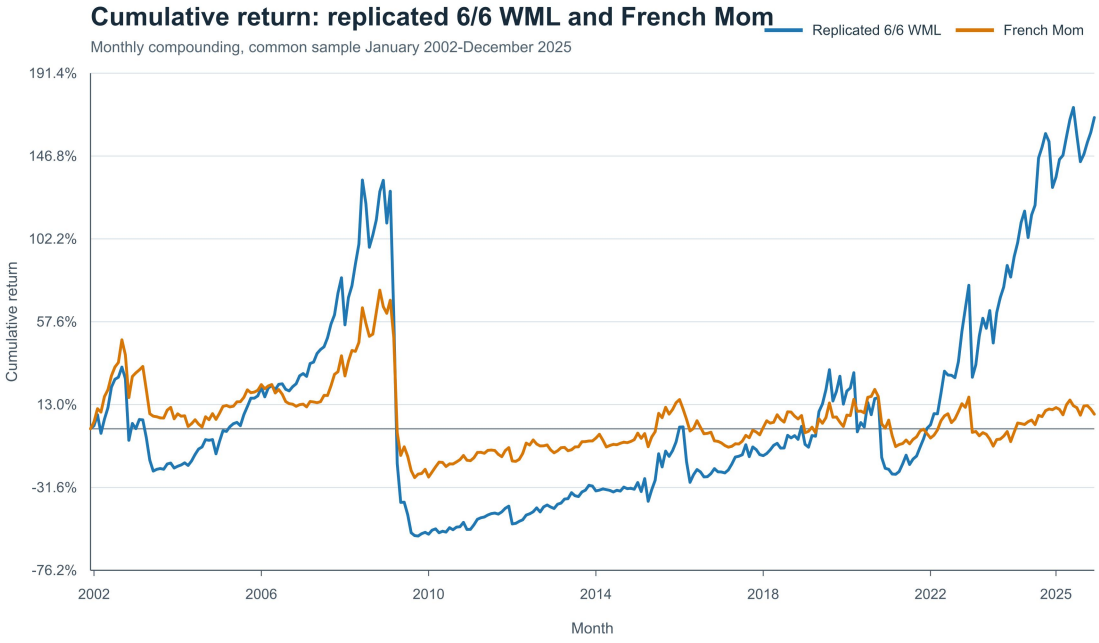
Monthly matched observations: N=288, January 2002–December 2025. JT WML is the Panel A 6/6 strategy. Original factor percent units were converted to decimals before analysis.

Table 5.2 Independent Extension: Factor Correlation and OLS Regression

Parameter	Estimate	p-value
Pearson correlation	0.821	
Regression alpha (%/month)	0.476	0.0579
Regression beta	1.360	1.12e-71
R-squared	0.675	

Model: JT WML = $\alpha + \beta \times Mom + \varepsilon$. Alpha is not significant at 5% ($p \approx 0.058$).

Figure 1 Cumulative Returns: JT Momentum vs. Fama–French Mom (2002–2025)



6. Validation, Limitations, and Conclusion

6.1 Validation

Several validation procedures were conducted to assess the reliability of the replication.First, the original data were inspected for missing observations, duplicate stock-month records, unusual returns, and security classifications. The sample-construction process was documented and implemented without modifying the raw datasets.

Second, the Table I portfolio construction was independently validated. The validation checks confirmed the accuracy of momentum rankings, decile assignments, overlapping holding periods, portfolio weights, and winner-minus-loser return calculations. The monthly strategy results were independently reconstructed without numerical discrepancies.

Additional consistency checks confirmed that the full-sample momentum returns used in Tables II and III matched the corresponding 6/6 strategy from Table I.

6.2 Limitations

The study has several limitations. First, the modern sample covers 2000–2025 rather than the original 1965–1989 period. Differences in market structure, trading conditions, and stock characteristics may contribute to the differences in empirical results.

Second, the treatment of unavailable holding-period returns may influence portfolio performance. The implementation retains available delisting-adjusted returns but reweights remaining securities when subsequent returns are unavailable, which may introduce selection bias.

Third, the one-week gap is implemented using seven calendar days because the original paper does not specify the exact trading-day convention.

Finally, the reported returns do not fully incorporate transaction costs, short-selling constraints, or financing costs, which may reduce the profitability of momentum strategies in practical trading.

6.3 Conclusion

This project conducts a modern-sample replication of Jegadeesh and Titman (1993) using US stock market data from 2000 to 2025.

The results show that all 32 momentum strategies generate positive average returns, although their raw returns are generally less statistically significant than those reported in the original paper. Further analysis reveals negative market beta exposure for the winner-minus-loser strategy and significant CAPM-adjusted abnormal returns.

The independent extension finds that the replicated JT momentum strategy is strongly correlated with the Fama–French Momentum Factor, suggesting that both strategies capture similar return patterns despite their different construction methods.

Overall, the findings provide evidence of momentum-related return continuation in the modern US equity market, while highlighting the importance of statistical uncertainty, portfolio construction choices, and implementation costs when evaluating momentum strategies as practical investment tools.